

Draft - expert review pending

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Unit 3 assessment

Unit summary

A pattern grows by a rule (fixed step, doubling, or repeating), and the growth type points to the rule's shape. A variable names a quantity that varies; an expression is a machine turning each input into its output, built from variables, coefficients and terms. An equation is a balance: legal moves do the same thing to both sides, and the method is simplify, isolate the variable term, isolate the variable, check. An inequality claims a range; multiplying or dividing by a negative flips the sign because the number line turns around. Algebra solves daily problems through a loop: translate the situation into symbols, solve, interpret back into the situation, and decide - and comparing two expressions settles a whole family of questions at once.

Summative assessment

Multiple-choice questions (MCQs)

1. The first five terms of a sequence that starts at 7 and grows by a step of 4 are: (*Remember*) a) 7, 11, 15, 19, 23 b) 7, 10, 13, 16, 19 c) 4, 7, 11, 15, 19 d) 7, 14, 21, 28, 35
2. For the matchstick rule $2t + 1$, the sticks needed for 12 triangles are: (*Apply*) a) 24 b) 25 c) 36 d) 13
3. In the expression $500 + 30k$, the coefficient is: (*Understand*) a) 500 b) 30 c) k d) 530
4. The expression $2t + 1$ is best described as: (*Understand*) a) an equation waiting to be solved b) a rule that turns any number of triangles into its

stick count c) a pair of numbers added together d) an inequality in disguise

5. The solution of $30 + 12m = 150$ is: (*Apply*) a) $m = 5$ b) $m = 10$ c) $m = 12$ d) $m = 15$
6. The solution of $5x - 7 > 18$ is: (*Apply*) a) $x > 5$ b) $x < 5$ c) $x > 25$ d) $x < 25$
7. Solving $-3y \leq 21$ gives: (*Apply*) a) $y \leq -7$ b) $y \geq -7$ c) $y \leq 7$ d) $y \geq 7$
8. A photocopy shop charges 30 rupees to open the machine plus 8 rupees per copy. The cost of 25 copies is: (*Apply*) a) 200 rupees b) 230 rupees c) 254 rupees d) 158 rupees
9. Van A costs $500 + 30k$ rupees and Van B costs $300 + 40k$ rupees for k kilometres. The two cost the same at: (*Apply*) a) $k = 10$ b) $k = 20$ c) $k = 25$ d) $k = 50$
10. The four moves of algebraic problem solving, in order, are: (*Understand*) a) solve, translate, decide, interpret b) translate, solve, interpret, decide c) interpret, translate, solve, decide d) translate, interpret, solve, decide

Restricted-response questions (RRQs)

1. Write the first five terms of a sequence that starts at 12 and grows by a step of 5. (*Apply*)
2. For the matchstick rule $2t + 1$, compute the sticks for 12 triangles, and explain in one sentence what the $+ 1$ accounts for. (*Apply*)
3. In the expression $45g + 20$ (juice glasses at 45 rupees each plus a 20-rupee box charge), name the variable, the coefficient, and state what the constant 20 represents. (*Understand*)
4. Solve $60 + 15k = 195$ and prove the answer by substitution. (*Apply*)
5. Write and solve the equation for: a tutor charges 250 rupees for materials plus 400 rupees per session, and a family paid 2,650 rupees in total. (*Apply*)
6. Solve $-4p > 24$ and explain in one sentence why the inequality sign flips. (*Apply*)
7. A trainee writes $60 + 15k = 195 = 135 = 9$ on a single line. Name the misconception in one sentence. (*Analyze*)

8. Name the four moves of algebraic problem solving, and state which move the substitution check belongs to. (*Understand*)
9. Printer A costs $2,000 + 15n$ rupees and Printer B costs $1,200 + 25n$ rupees for n programmes. Find the value of n at which both cost the same. (*Apply*)
10. A friend says "algebra is separate from real maths". Correct the claim in two sentences using the rate-card idea. (*Analyze*)

Extended-response questions (ERQs)

1. A decorator hangs bunting in triangles along the school gate: one triangle needs 3 metres of ribbon, each extra joined triangle adds 2 metres, and there is a fixed 5-metre tail at each end of the string. Write the ribbon expression for n triangles, compute the ribbon for 15 triangles, and state the largest whole number of triangles a 70-metre ribbon can carry, with the substitution check. (*Analyze*)
2. A school bus service charges 8,000 rupees per trip plus 120 rupees per pupil, and the transport budget for one event is 20,000 rupees. Write the cost expression for p pupils, write and solve the inequality for how many pupils the budget can carry, and state the largest whole number of pupils with the check that proves one more pupil would not fit. (*Analyze*)
3. A school event needs printed programmes. Printer A charges 2,000 rupees setup plus 15 rupees per programme; Printer B charges 1,200 rupees setup plus 25 rupees per programme. Write both cost expressions, compute both totals for 150 programmes and state which is cheaper, and find the number of programmes at which both cost the same; then recommend which printer the school should choose if the print run might grow or shrink, with a reason. (*Analyze*)
4. A school can spend at most 2,000 rupees on geometry boxes costing 45 rupees each, and must first pay a 300-rupee debt to the supplier from the same money. Write and solve the inequality for the number of boxes, state the largest whole number of boxes with the substitution check, and write an expression for the money left over after buying that many boxes. (*Analyze*)
5. Using the real rate card you collected at your practicum school (Topic 3.3 practicum task), write its cost expression, compute the cost for three realistic quantities, and compare it with a flat-rate alternative: find the quantity at which the two cost the same, and recommend which option the school should prefer, with reasons. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. **a** - start at 7 and add 4 each time: 7, 11, 15, 19, 23.
2. **b** - $2 \times 12 + 1 = 25$.
3. **b** - the coefficient 30 multiplies the variable k .
4. **b** - an expression is a rule or machine; nothing is claimed equal.
5. **b** - $150 - 30 = 120$; $120 / 12 = 10$.
6. **a** - $5x > 25$, so $x > 5$.
7. **b** - dividing by -3 flips the sign: $y \geq -7$.
8. **b** - $30 + 8 \times 25 = 30 + 200 = 230$.
9. **b** - $500 + 30k = 300 + 40k$ gives $200 = 10k$, so $k = 20$.
10. **b** - translate, solve, interpret, decide.

RRQ model answers and mark schemes

1. Model: 12, 17, 22, 27, 32. **Marks (2):** 1 - first three correct; 1 - all five.
2. Model: 25 sticks; the + 1 is the starting stick of the first triangle that no later triangle shares. **Marks (2):** 1 - 25; 1 - a sensible account of the constant.
3. Model: g is the variable (glasses), 45 is the coefficient (price per glass), 20 is the fixed box charge. **Marks (3):** 1 - variable; 1 - coefficient; 1 - constant's meaning.
4. Model: $15k = 135$; $k = 9$; check $60 + 15 \times 9 = 195$. **Marks (2):** 1 - $k = 9$; 1 - substitution check shown.
5. Model: $250 + 400s = 2,650$; $400s = 2,400$; $s = 6$ sessions. **Marks (2):** 1 - correct equation; 1 - solution.
6. Model: $p < -6$ (dividing by -4 flips the sign); the flip happens because multiplying or dividing by a negative turns the number line around. **Marks (2):** 1 - $p < -6$; 1 - the number-line explanation.
7. Model: the equals sign is being used to mean "the answer comes next"; each step should be on its own line because the equals sign claims balance, and 195 does not equal 135. **Marks (2):** 1 - names the misconception; 1 - states the correct practice.

8. Model: translate, solve, interpret, decide; the substitution check belongs to the solve move (it proves the solution). **Marks (3):** 1 - four moves in order; 1 - check assigned to solve; 1 - a sentence on why.
9. Model: $2,000 + 15n = 1,200 + 25n$; $800 = 10n$; $n = 80$. **Marks (2):** 1 - equation set up; 1 - $n = 80$.
10. Model: algebra is the compression of daily arithmetic into reusable rules; the rate card, the fare and the budget are the same expression in different clothes, so algebra is exactly the maths of real situations. **Marks (2):** 1 - the compression idea; 1 - the rate-card example or equivalent.

ERQ rubrics

1. Bunting. Analytic rubric: *Expression* (0-2: $2n + 11$ built with parts labelled); *Computation* (0-2: 41 metres for 15 triangles); *Largest n with check* (0-3, **Analyze:** $2n + 11 \leq 70$ gives $n \leq 29.5$, so 29 triangles, with $2(29) + 11 = 69 \leq 70$ shown). Total 7.
2. Bus budget. Analytic rubric: *Expression* (0-2: $8,000 + 120p$); *Inequality solved* (0-2: $p \leq 100$); *Largest whole number with check* (0-3, **Analyze:** 100 pupils, with $8,000 + 120 \times 100 = 20,000$ exactly and 101 pupils shown to exceed at 20,120). Total 7.
3. Printers. Analytic rubric: *Expressions* (0-2: $A = 2,000 + 15n$, $B = 1,200 + 25n$); *Totals and comparison* (0-2: 4,250 vs 4,950, A cheaper by 700); *Break-even and recommendation* (0-3, **Analyze:** $n = 80$ with the working, and a recommendation that weighs a possibly smaller run, below which B is cheaper, with a reason). Total 7.
4. Geometry boxes. Analytic rubric: *Inequality* (0-2: $45b \leq 1,700$); *Solution and largest whole number* (0-2: $b \leq 37.8$, so 37 boxes); *Check* (0-2: $37 \times 45 = 1,665 \leq 1,700$ while $38 \times 45 = 1,710$ exceeds); *Leftover expression* (0-2, **Analyze:** $1,700 - 45b$, or 35 rupees left at $b = 37$). Total 8.
5. Practicum rate card. Analytic rubric: *Expression* (0-2: a correct expression from the real card); *Three quantities* (0-2: three correct computations); *Break-even* (0-3, **Analyze:** the flat-rate comparison set up and solved, with the quantity where the two costs cross); *Recommendation* (0-2: justified against the school's likely usage); *Clarity* (0-1). Total 10.

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